Sanskar Jain

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Summary

Systems-oriented backend engineer with 2+ years building distributed, cloud-native applications at Verisk Analytics. Strong in **database technologies** (MongoDB, PostgreSQL), **Linux**-based containerised environments, and async **inter-process communication** patterns using Kafka, SQS, and SNS. Delivered fault-tolerant, scalable systems end-to-end with a record of independent ownership, clear technical communication, and zero-downtime production deployments.

Skills

Languages: C#, Python, Java, C++ (academic)
Backend: .NET 8, ASP.NET Core, Spring Boot, REST APIs, Microservices, Kafka, Redis, CQRS
Messaging & IPC: Kafka, AWS SQS/SNS, Redis Pub/Sub, Event-Driven Architecture, DLQ, Retry/Backoff
Platforms & DevOps: Linux, AWS (S3, EKS, ECS, Lambda, ElastiCache), Docker, Kubernetes, Terraform, CI/CD
Databases: MongoDB, PostgreSQL, MySQL, Indexing, Query Optimisation, Migration Scripting
Tools: OAuth 2.0, JWT, Okta, Kong Gateway, OpenTelemetry, Splunk, Swagger/OpenAPI, Jira

Experience

Verisk Analytics Jan 2024 – Present
Software Engineer II, Hyderabad, India

- Co-designed **RulesRunner**, a stateless microservice using **CQRS** and **AWS SQS** for async inter-process communication, offloading compliance checks and reducing latency from **3000ms to 500ms (83%)** with horizontal scaling.
- Architected event-driven pipeline with **idempotent processing**, retry/backoff strategies, DLQ handling, and **AWS Lambda** triggers – ensuring zero message loss across distributed processes under failure conditions.
- Authored technical design documentation defining service contracts, IPC patterns via **SNS** fan-out, and DLQ strategies; aligned engineering and product stakeholders on distributed system behaviour.
- Engineered scalable backend for **Xactimate's** import/export of **30+ data types** using async processing, **S3 pre-signed URLs**, and **MongoDB**, reducing transfer time by **38%** and data issues by **25%**.
- Built REST APIs on **Linux-based AWS EKS** clusters handling **200K+ weekly requests** with structured error handling, input validation, and **Swagger/OpenAPI** documentation.
- Established **OAuth 2.0** authentication behind **Kong Gateway** on **AWS EKS**, integrating **Okta JWT** tokens with Identity Server for secure cross-service authorisation.
- Improved API response time by **34%** via **Redis/ElastiCache** caching, connection pooling, and query optimisation on high-traffic endpoints.
- Optimised **database indexing strategies** on **MongoDB** and **PostgreSQL** by analysing slow-query logs and access patterns, reducing p95 query latency across critical workflows.
- Authored **production-safe database migration scripts** with idempotent operations and full backward compatibility, enabling zero-downtime deployments.
- Slashed deployment time by **50%** and build cycles by **33%** by overhauling **CI/CD pipelines** with staged gates, parallel jobs, and automated test integration on Linux build agents.

Projects

Expense Tracker – Distributed Microservices Application Aug 2023 – Dec 2023

- Built **5+ microservices** communicating via async **Kafka** messaging (IPC), with **Kong API Gateway** handling routing, auth, and load balancing across services.
- Containerised all services with **Docker** on Linux and orchestrated deployments using **Kubernetes**; simulated **1000+ transactions** end-to-end for load and fault-tolerance testing.

Certifications

AWS Certified Cloud Practitioner Amazon Web Services — Oct 2024

Education

B.Tech in Computer Engineering – Vishwakarma Institute of Technology, Pune 2024 — CGPA: 8.8